Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/18/2024
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Investigating Long-term Structural and Functional Changes in the Eye and Brain After Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brandon.r.macias@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Johnson Space Center Cardiovascular and Vision Laboratory		
PI Address 2:	2101 NASA Parkway, HAC/B21N-1207		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	02/15/2021	**End Date:**	09/30/2033
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2033 per C. Ribeiro/JSC (Ed., 3/21/23). NOTE: End date changed to 02/14/2023 per C. Ribeiro/JSC (Ed., 4/7/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Huang, Alex M.D., Ph.D. (Doheny Eye Institute) Karanjia, Rustum M.D., Ph.D. (Doheny Eye Institute) Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Martin, Bryn Ph.D. (Alcyone Therapeutics Inc.) Sadda, Srinivas M.D. (Doheny Eye Institute) Young, Millennia Ph.D. (NASA Johnson Space Center)		

Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The proposed study will identify long-term health consequences of spaceflight on the structure and function of the eye and brain. Experiments will be performed with crewmembers before, during, and after six-month spaceflight missions; post-flight surveillance will span five years. In another arm we will characterize the structural and functional changes to the eye and brain, including cognitive changes, in crewmembers who have participated in past missions, calling them back several years after their most recent flights. The data delivered from this study will provide NASA with information necessary for completing the risk assessment for future interplanetary expeditions. This proposal primarily addresses the following two Human Research Roadmap Gaps: 1) We do not know the etiological mechanisms and contributing risk factors for ocular structural and functional changes seen in-flight and post-flight and 2) Determine if exposure to long-duration spaceflight leads to neural structural alterations and if this remodeling impacts cognitive and functional performance. Ed. Note (2/29/24): This project has been combined with the work of three other Principal Investigators (PIs) who responded to the same solicitation with independent proposals. The protocols and aims from all four PIs have been integrated into a revised proposal that was delivered to NASA Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP) Element Scientists in January 2022. For information on the four related investigations, see: • The Long-Term Consequences of Spaceflight on Brain and Eye Health (PI: Bershad). • Long-Term Brain Structural and Functional Consequences of Spaceflight (PI: Basner). • Investigating Long-term Structural and Functional Changes in the Eye and Brain After Spaceflight (PI: Macias). • Recovery Timeline of Spaceflight-Induced Central Nervous System Changes (PI: Seidler).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Characterizing the long-term effects of Spaceflight Associated Neuro-ocular Syndrome (SANS) will provide NASA with information necessary for completing the risk assessment for future interplanetary expeditions. Furthermore, the unique nature of this project may lead to the adoption of new hardware, software, or protocol elements in clinical practice, benefiting patients on Earth. For example, novel methods to be used in this study are augmenting capabilities of existing technologies, e.g., deformation mapping for ocular Magnetic Resonance Imaging (MRI).
Task Progress:	PROGRESS DURING THIS REPORTING PERIOD This “Long-Term Health” (LTH) Virtual NASA Specialized Center of Research (VNSCOR) has a ground and flight component, and therefore two Research Operations and Integration (ROI) coordination teams. Dr. Macias’ Principal Investigator (PI) team is supporting bi-weekly meetings with ROI to implement both portions of the projects. In addition, Dr. Macias’ PI team is holding bi-weekly meetings with the LTH Co-PI teams to coordinate the overall implementation effort of the ground and flight components of the project. The “LTH Call Back” retrospective arm will focus on investigating the structural and functional changes in the eye, brain, and cognition in astronauts who previously participated in long-duration missions, calling them back at least 5 years after their previous flights to determine changes that may endure long after return from spaceflight. Major accomplishments during this reporting period for this sub-study include: • The Lifetime Surveillance of Astronaut Health (LSAH) data share request was approved by the LSAH Advisory Board. • An LSAH epidemiologist was added to the study team to help with data analysis. • The MRI facility contract was finalized, and two MRI dry runs were completed at Houston Physicians’ Hospital. • One internal dry run was completed in the Cardiovascular and Vision Laboratory. • The neuropsychological and neurological Co-PI team completed a dry run at NASA Johnson Space Center (JSC). • The Crew Office approved the Informed Consent Briefing (ICB). • Recruitment efforts were initiated by ROI and are ongoing. • The Test Readiness Review package was developed and is undergoing internal review. Ongoing forward work with ROI for “LTH Call Back” includes: • Approval of the TRR package (planned on December 2024). • Initiate recruitment and testing of astronauts and matched control TSS participants. The “Brain Health” component is prospective and will investigate the development and recovery of structural and functional changes in the eye, brain, and cognition in astronauts participating in standard or extended duration missions and will follow them out 5 years postflight. Major accomplishments for the Flight Study during this reporting period include: • ROI submitted the Feasibility Assessment to the Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP) Elements, and the sub-study was approved for Select for Flight in December 2024. • The LSAH data share request was approved by the LSAH Advisory Board. • An LSAH epidemiologist was added to the study team to help with data analysis. Ongoing future work with ROI for “Brain Health” includes (pending selection for flight determination by NASA's Human Research Program/HRP): • Completion of two MRI dry runs for the Brain Health MRI scanning protocol. • Development and approval of a TRR package (targeting January 2025). • Development and approval of ICB presentation. • Pitch ICB starting with Crew-12.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Carter KJ, Basner M, Bershad EM, Seidler R, Stahn AC, Laurie SS, Macias BR. "Investigating long-term structural and functional changes to the eye and brain after spaceflight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024